VISVESVARAYA TECHNOLOGICAL UNIVERSITY

"JNANA SANGAMA",MACHHE, BELAGAVI-590018

Activity

on

CLOUD DATA ENGINEERING (AI371)

Quiz on Module 2

Submitted in partial fulfillment of the requirements for the VI semester

Bachelor of Engineering

in

Artificial Intelligence and Machine Learning

of

Visvesvaraya Technological University, Belagavi

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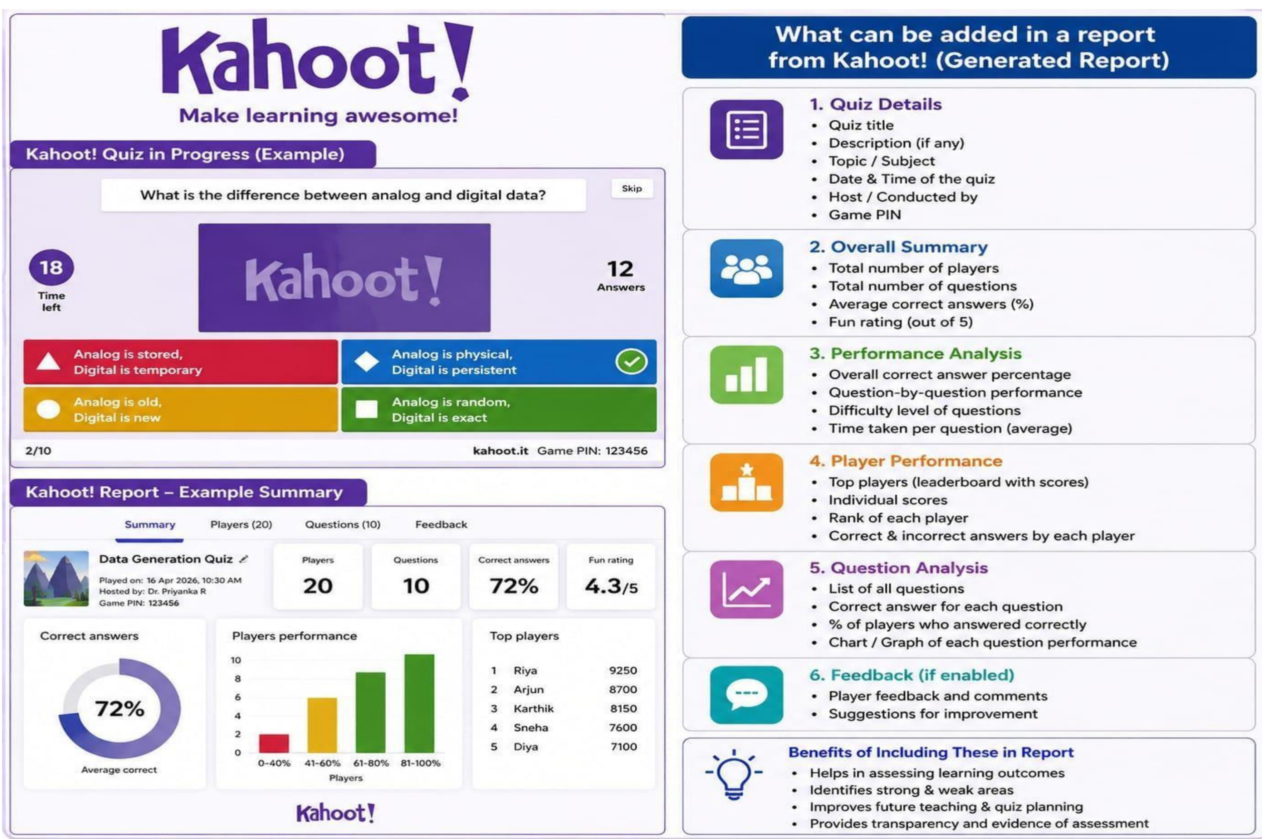
INTRODUCTION

Cloud Data Engineering focuses on designing, building, and managing data pipelines and architectures using cloud platforms. It involves working with large-scale data systems to collect, store, process, and analyze data efficiently. With the growth of big data and digital transformation, cloud-based tools and services enable scalable, flexible, and cost-effective data solutions. A quiz on cloud data engineering helps assess foundational knowledge of concepts such as data pipelines, ETL (Extract, Transform, Load), cloud storage, distributed computing, and modern data platforms.



OBJECTIVE

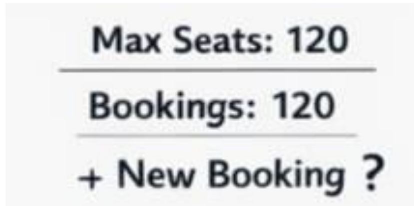
The objective of a cloud data engineering quiz is to evaluate understanding of key concepts, tools, and practices used in handling data in cloud environments. It aims to test knowledge of data processing frameworks, cloud services, data warehousing, real-time data streaming, and data security. Additionally, it helps learners identify their strengths and areas for improvement while reinforcing theoretical and practical aspects of cloud-based data engineering.

MCQ QUESTIONS WITH ANSWERS AND EXPLANATION

1. Which type of data is JSON?
A) Structured
B) Unstructured
C) Semi-structured
D) Binary
Correct option: Semi-structured
JSON (JavaScript Object Notation) is considered semi-structured data because it does not follow a strict tabular schema like structured data. However, it still maintains an organized format using key–value pairs. This makes it flexible and easy to read and write.
2. What is the difference between analog and digital data?
A) Analog is stored, digital is temporary
B) Both same
C) Analog is physical, digital is persistent
D) None
Correct answer: Analog is physical, digital is persistent
Analog data represents information in a continuous physical form, such as sound waves, temperature, or voltage signals. It exists naturally and varies smoothly over time. Digital data is more persistent because it can be stored for long periods without degradation. Hence, analog is physical while digital is persistent in nature.
3. What happens if step 2 fails?

A) Step 1 is saved
B) Entire process is rolled back
C) Step 3 is ignored
D) Partial update occurs
Correct Answer: B) Entire process is rolled back
If step 2 fails, the system typically follows the concept of a transaction where all steps must succeed together. In such cases, a failure in any step causes the entire process to be reversed to maintain consistency. This is known as a rollback, ensuring no partial changes are saved. It helps preserve data integrity and prevents incomplete or incorrect results.

4. System Should:

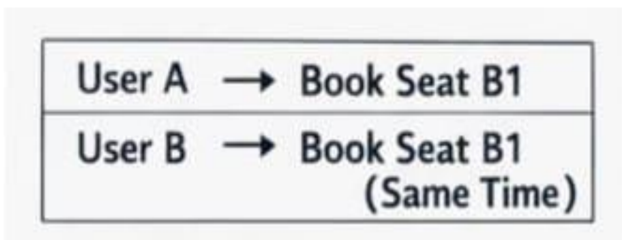


A) Allow booking
B) Overbook
C) Reject booking
D) Ignore rule

Correct Answer: C) Reject booking

Ignoring the rule means the system does not apply or enforce any booking constraints or validation checks while processing a request. Even if the booking conditions are not satisfied, the system continues without verifying the rules. This results in improper handling of data and can cause inconsistent or unreliable behavior.

5. Result?

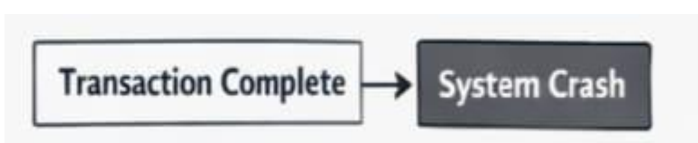


A) Both succeed
B) Both fail
C) One succeeds, one fails
D) Duplicate booking

Correct Answer: C) One succeeds, one fails

In such scenarios, two operations are attempted simultaneously but only one completes successfully. The other fails due to conflicts like resource unavailability or system constraints. This ensures that rules (like avoiding duplicates) are maintained. It prevents both from succeeding when only one is allowed.

6. Data will be:

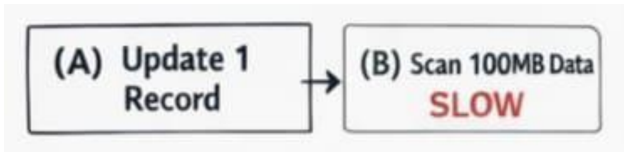


A) Lost
B) Permanently stored
C) Temporary
D) Corrupted

Correct Option: B) Permanently stored

When data is saved to a storage medium like a hard disk, SSD, or database, it becomes permanently stored. This means the data remains intact even after the system is turned off or restarted. Permanent storage ensures that information can be retrieved and used later whenever needed.

7. System(A) is:



A) Update 1 Record

B) Data SLOW

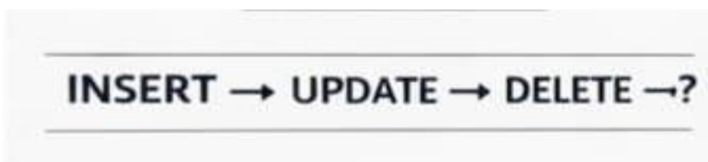
C) Graph DB

D) File system

Correct Option: B) Data SLOW

A system described as "Data SLOW" typically indicates poor performance in handling data operations such as reading, writing, or updating records. This means the system takes more time to process requests, which can affect user experience and efficiency. Slow data systems may result from inefficient algorithms, lack of indexing, or outdated hardware.

8. What is Captured?



A) insert

B) All changes

C) Only delete

D) None

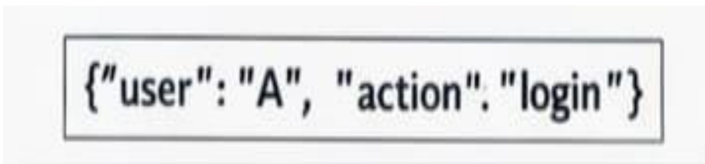
Correct Option: B) All changes

"Captured" refers to recording or tracking every modification made to data within a system. This includes inserts, updates, and deletions, ensuring a complete history of changes. Capturing all changes is important for auditing, recovery, and maintaining data integrity. It allows systems to monitor what has been modified over time.

9. This is :

A) Table
B) API response
C) Log entry
D) Query
Correct Option: C) Log entry
A log entry is a record that captures events or actions occurring within a system. It typically includes details like timestamps, operation type, and status of the event. Logs are used for monitoring, debugging, and auditing system activities.

10. Log type?

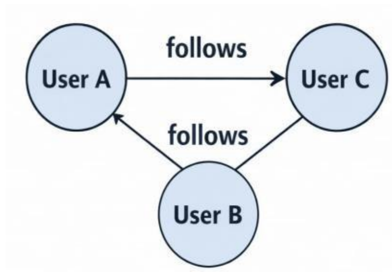


A) Binary
B) Plain text
C) Semi-structured
D) Unstructured
Correct Option: C) Semi-structured
Logs are typically considered semi-structured because they follow a consistent format but are not strictly organized like tables. They may include fields such as timestamps, log levels, and messages, but the content can vary. This flexibility allows logs to capture different types of events without a rigid schema.

11. This represents:

A) Update
B) Delete
C) Insert-only
D) OLAP
Correct Option: C) Insert-only
"Insert-only" represents a system where data is only added and never modified or deleted. Each new record is appended, preserving all historical information. This approach is useful for audit trails, logging, and event tracking because it ensures no data is lost. Unlike update or delete operations, existing records remain unchanged. Hence, it clearly represents an insert-only model.

12. Data model?



A) Relation
B) Files
C) Graph
D) OLTP
Correct Option: C) Graph
A graph data model represents data as nodes (entities) and edges (relationships) between them. It is especially useful for handling highly connected data such as social networks, recommendation systems, or network analysis. Unlike relational models, it focuses on relationships rather than tables and rows.
13. Type of data?

Time →	10:00	10:01	10:02
Value →	28	29	30

A) Graph
B) Binary
C) Time-series
D) API
Correct Option: C) Time-series
Time-series data represents values recorded at specific time intervals, such as logs, sensor readings, or stock prices. It focuses on how data changes over time, making time an essential component of each record. This type of data is widely used for monitoring, forecasting, and trend analysis. Unlike graph or binary data, time-series emphasizes chronological order. Hence, the given data type is time-series.
14. Replica is for:

A) EC2
B) Speed
C) Processing
D) Backup

Correct Option: D) Backup

A replica is a copy of data stored in another location to ensure safety and availability. Its primary purpose is to act as a backup in case the original data is lost, corrupted, or the system fails. Replication helps in disaster recovery by allowing data to be restored quickly. It also ensures data durability across systems. Hence, replicas are mainly used for backup purposes.

15. System responds even if data not updated.What is the property?

A) Atomicity

B) Consistency

C) Isolation

D) Basically Available

Correct Option: D) Basically Available

"Basically Available" is a property from the BASE model used in distributed systems. It means the system continues to respond to requests even if some data may not be fully updated or consistent. The focus is on high availability rather than immediate correctness.

16. OLTP systems are designed for large-scale data analysis.

A) True

B) False

Correct Answer: B) False

OLTP (Online Transaction Processing) systems are designed for handling day-to-day transactional operations like insert, update, and delete. They focus on fast query processing and maintaining data consistency in real time. OLTP systems are not meant for large-scale data analysis; that role is handled by OLAP systems.

17. JSON is a semi-structured data format.

A) True

B) False

Correct Answer: A) True

JSON (JavaScript Object Notation) is considered a semi-structured data format because it does not follow a rigid schema like relational databases. It organizes data in key-value pairs, which allows flexibility in storing and representing information. The structure is consistent enough to be machine-readable but still adaptable to different types of data. This makes JSON widely used in APIs and web applications.

18. Identify mapping?

POST → Create
GET → Read
PUT → Update
DELETE → Delete

A) CRUD-HTTP

B) Graph

C) API logs

D) OLAP

Correct Option: A) CRUD-HTTP

CRUD operations (Create, Read, Update, Delete) are commonly mapped to HTTP methods in web systems. For example, Create → POST, Read → GET, Update → PUT/PATCH, and Delete → DELETE. This mapping helps web applications interact with databases through APIs in a standardized way. It ensures clear communication between client and server using HTTP protocols.

19. Block storage network

A) SAN

B) OLTP

C) EC2

D) Graph

Correct Answer: A) SAN

SAN (Storage Area Network) is a high-performance network designed specifically for block-level storage. It provides direct access to storage devices, making it suitable for applications that require fast and reliable data access. In SAN, storage appears to the system as if it is locally attached disk storage. This allows multiple servers to access shared storage efficiently.

20. Binary logs are human-readable

A) True

B) False

Correct Answer: B) False

Binary logs are not human-readable because they are stored in a compact binary format instead of plain text. They are designed for efficient storage and fast processing by machines, not for direct reading by users. These logs contain encoded information about database changes such as inserts, updates, and deletes. To view their content, special tools or commands are required to decode them. This improves performance and reduces storage space but sacrifices readability.

21. Virtual machine in cloud

A) EBS

B) EC2

C) SAN

D) CDC

Correct Answer: B) EC2

EC2 (Elastic Compute Cloud) is a service that provides virtual machines in the cloud. It allows users to run applications on scalable computing resources without needing physical hardware. Each EC2 instance behaves like a virtual server with configurable CPU, memory, and storage. It is widely used for hosting applications, running workloads, and development tasks. Unlike EBS (storage), SAN (storage network), or CDC (data replication concept), EC2 provides actual compute power.

22. Eventually consistent data
A) ACID
B) CRUD
C) BASE
D) OLAP
Correct Answer: C) BASE
BASE stands for Basically Available, Soft state, and Eventually consistent, and it is commonly used in distributed systems. In eventual consistency, the system does not guarantee immediate consistency across all nodes. Instead, it allows temporary differences in data, which will be synchronized over time. This approach improves availability and scalability in large distributed systems. Unlike ACID, which focuses on strict consistency, BASE prioritizes performance and availability. CRUD refers to operations, and OLAP is for analysis, not consistency models.
23. Temporary inconsistent data
A) Strong
B) Durable
C) Atomic
D) Soft State
Correct Answer: D) Soft State
Soft State refers to a condition in distributed systems where data may be temporarily inconsistent. It allows the system to change over time even without new input due to eventual synchronization. This means different nodes may not have the same data at all times. However, the system will gradually reach consistency through updates.
24. Parallel servers storing data
A) Slow
B) Scalable
C) Isolated
D)None
Correct Answer: B) Scalable Parallel servers storing data indicate a system designed to handle increasing workloads efficiently. When multiple servers work together, data processing and storage capacity can
be expanded easily. This improves performance without overloading a single system. Such architecture supports horizontal scaling by adding more servers as needed.
25. What are challenges in APIs?
A) Fixed schema
B) High storage
C) No latency
D) Authentication
Correct Answer: D) Authentication

One of the major challenges in APIs is ensuring proper authentication and security. APIs often expose sensitive data and services, so verifying the identity of users or systems is essential. Without strong authentication, unauthorized access can lead to data breaches and misuse. Common methods include API keys, tokens, and OAuth mechanisms. Managing authentication across multiple services can become complex in distributed systems.

26. ATM with drawal persists even after crash

A) Atomicity

B) Isolation

C) Consistency

D) Durability

Correct Answer: D) Durability

Durability ensures that once a transaction is successfully completed, its changes are permanently saved in the system. In the case of ATM withdrawal, once the transaction is confirmed, the deducted amount must not be lost even if a system crash occurs.The updated account balance is stored in non-volatile storage like a database or disk. This guarantees that the transaction result remains intact after failures or restarts.

27. CDC captures:

A) Only insert

B) Only delete

C) Insert,Update,Delete

D) Only read

Correct Answer: C) Insert, Update, Delete

CDC (Change Data Capture) is a technique used to track changes made in a database. It records all types of data modifications, including insertions, updates, and deletions. This helps in keeping systems synchronized by identifying exactly what has changed. CDC is widely used in data replication, ETL processes, and real-time analytics. It ensures that downstream systems receive only the modified data instead of full datasets.

28. An app fetches live weather data every minute from an external service

A) API

B) File-base

C) Graph DB

D) CSV

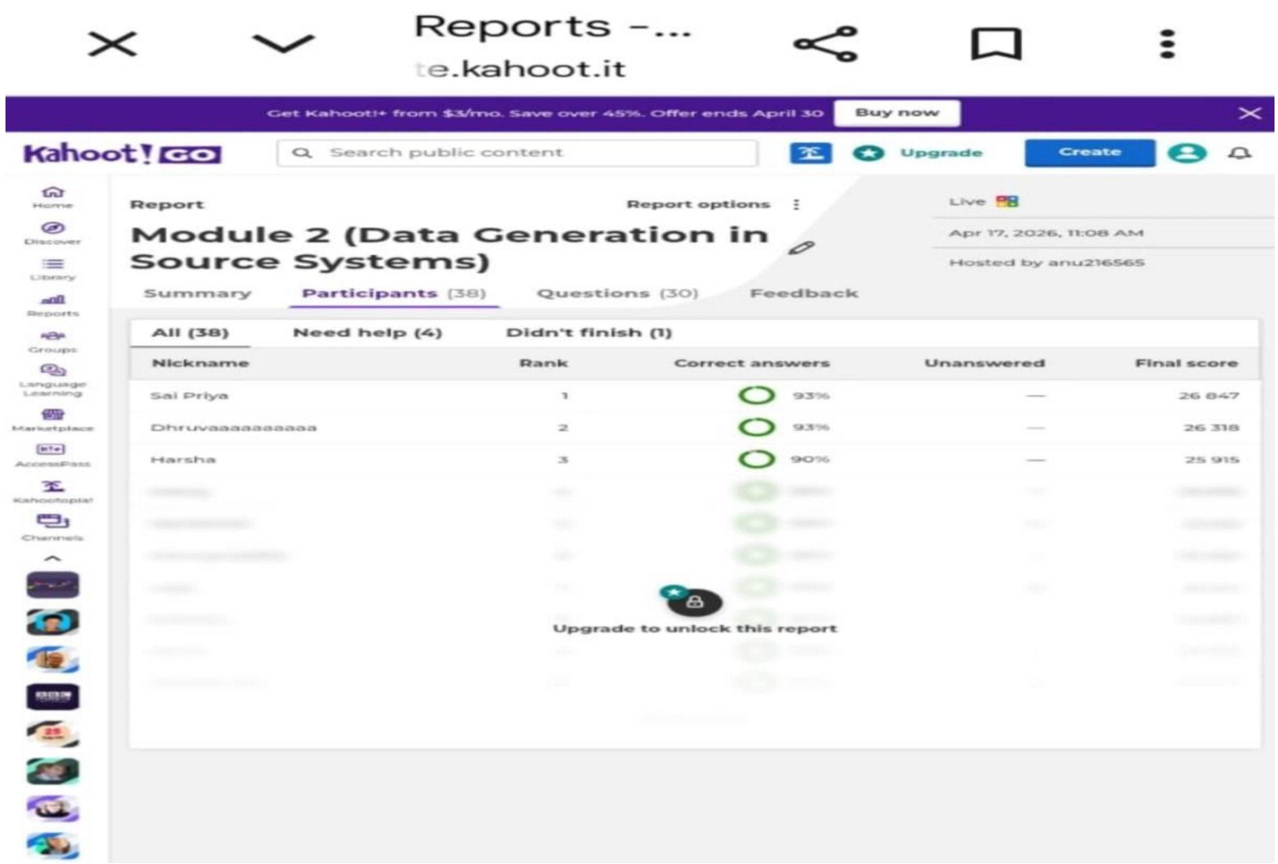
Correct Answer: A) API

An API (Application Programming Interface) allows an application to communicate with external services and fetch live data. In this case, the app is continuously requesting weather updates from a remote weather service.

APIs are designed for real-time data exchange over the internet. They enable structured communication between client and server using HTTP requests.

29. A banking system ensures balance never goes negative after transaction.
A) Atomicity
B) Isolation
C) Consistency
D) Durability
Correct Answer: C) Consistency
Consistency ensures that a database always moves from one valid state to another after a transaction. In a banking system, it guarantees that rules such as "balance cannot be negative" are never violated. If a transaction tries to withdraw more money than available, it will be rejected to maintain data integrity. This property ensures that all constraints and business rules are followed strictly. It is one of the key ACID properties in database systems.
30. Which system is most efficient?
A) CSV
B) Graph DB
C) Relational
D) API
Correct Answer: B) Graph DB
A Graph Database (Graph DB) is considered most efficient for handling highly connected data. It stores data as nodes and relationships, allowing direct traversal between connected records. This makes queries involving complex relationships much faster compared to relational or CSV systems. It avoids costly joins by linking data directly through edges. Graph DBs are widely used in social networks, recommendation engines, and fraud detection systems. CSV and relational models are less efficient for relationship-heavy queries.

Outcome of conducting this quiz:
Checks understanding
Students' knowledge of concepts like analog vs digital data is tested.
Immediate feedback
Participants instantly see whether their answers are correct or wrong.
Engagement increases
Interactive quizzes make learning more interesting and competitive.
Identifies weak areas
You can see which concepts students are confused about.
Performance ranking
Kahoot gives scores and leaderboards based on accuracy and speed.
Better retention
Students remember concepts more effectively through quizzes.



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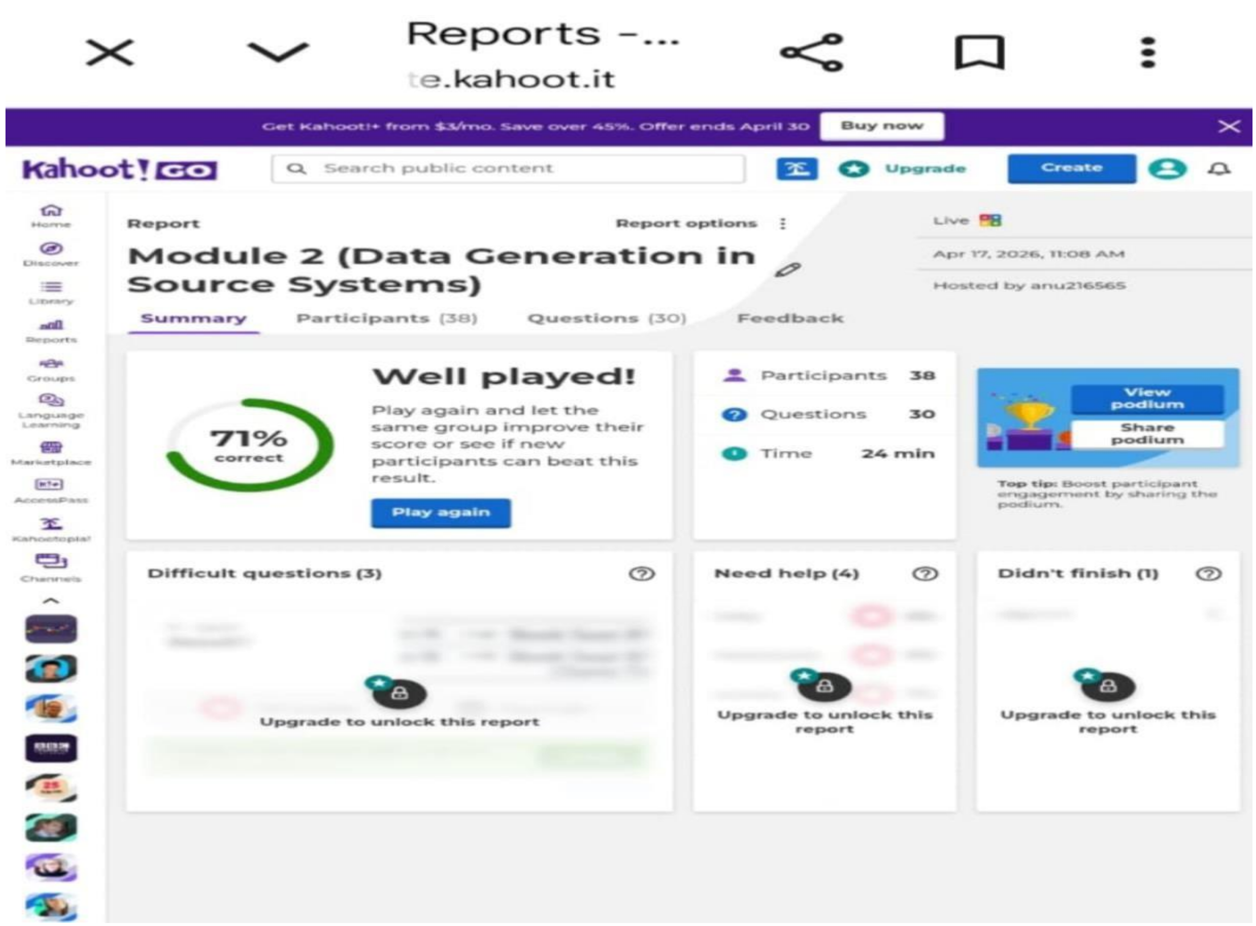

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CONCLUSION

In conclusion, a cloud data engineering quiz serves as an effective method to measure knowledge and enhance learning in this rapidly evolving field. It encourages a deeper understanding of cloud technologies and data handling techniques, preparing individuals for real-world applications and industry requirements. By engaging in such assessments, learners can strengthen their skills and stay updated with modern data engineering practices.